Inside the Rhino · Rhino's notes **@heyrhinotv**

one energy drink.

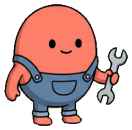
What one can does to your stomach, brain, heart and sleep, stop by stop. And who profits.

it's not energy. it's a loan.

Caffeine doesn't make energy. It borrows it from later, and you pay it back with interest. Call it the energy loan.

① stomach — 0–10 min

a whole day of sugar.



A typical can has about **40 g of sugar ≈ 10 teaspoons**. The American Heart Association's cap on added sugar is about **9 tsp a day for men and 6 for women**. One can beats it before dinner.

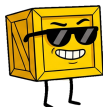
Caffeine moves fast: some is absorbed right in your stomach, the rest in your small intestine, and about **99% is in your blood within ~45 minutes**. Then it walks straight through the wall that guards your brain.

1 can > a whole day of sugar.

② brain + heart — 15–60 min

it hides tired. it doesn't add energy.





All day your brain piles up **adenosine**, a mail carrier with one message: "you're tired." Caffeine sits in its docks **without delivering the message**, so the messages pile up in the lobby. Peak caffeine: about **1 hour** in.

Your heart: in a small Mayo Clinic study (25 young adults, one 16-oz can), **blood pressure rose ~6%** vs ~1% with a placebo, and the stress hormone norepinephrine rose more than twice as much.

still hour one: the sugar rush is a myth.

31 studies, almost 1,300 adults: sugar didn't lift mood, and within an hour people felt **more tired and less alert**. The "energy blend" (taurine, ginseng, guarana) isn't backed by solid controlled trials, and extra B vitamins mostly leave in your pee. B2 is why it looks so yellow.

③ bedtime — 5 hrs

half is still in you.



Your liver clears it (one enzyme does over 90% of the first step). Caffeine's half-life is about **3–6 hours** for many adults. Call it 5. With a 200 mg can:

3 pm · 200 mg → 8 pm · ~100 → 1 am · ~50

In a sleep-lab study, **400 mg six hours before bed cut sleep by more than an hour**, and people didn't notice. When it clears, the stored-up tired walks in at once: the crash, with interest.

Your clock isn't my clock: smokers clear it faster (~6 h → 3.5 h), estrogen birth-control pills slow it by more than half, and late in pregnancy it can stretch to ~10.5 h.

④ the bill — who profits

the part that wakes you up costs less than a penny.



Bulk caffeine trades for about **$14 a kilogram**, so 200 mg costs about **$0.003**. You paid a few bucks.

The biggest energy drink company on the U.S. stock market sold about **$8.3 billion** of drinks last year, kept about **56¢ of every $1** after making and shipping them, then spent about **$2.2 billion** running the company: marketing, sponsorships, athletes, people.

Not a scam, a business. You're paying for the feeling, not the caffeine. *know what you're paying for.*

your receipt for one can.

10 minutes	a whole day of sugar
1 hour	peak caffeine, tired messages in the lobby, heart working harder
5 hours	half still in you
the caffeine	< ¼ of a penny
interest due	tonight

rhino's rule.

No caffeine within **6 hours of bed**, and treat that as the bare minimum. The researchers behind the sleep study said caffeine should generally be avoided **after 5 p.m.**

Need energy later than that? You're already in debt. Only sleep pays it off.

the only energy that doesn't come with interest.

read the label.

16-oz energy drinks range from **54 to 328 mg** of caffeine (FDA). About **400 mg a day** is usually fine for healthy adults; **seizures** can show up with ~1,200 mg taken fast.

ER visits involving energy drinks doubled to almost 21,000 a year (2007–2011); about 4 in 10 also involved other drugs or alcohol. Experts advise against energy drinks for kids and teens.

what's still in you tonight?

Plug in your can and your bedtime: Rhino's caffeine clock does the math for your body.

heyrhinotv.github.io/clock

Education and entertainment, not medical advice. Heart condition, pregnant, or on medication? Ask your doctor about caffeine. Half-life math is illustrative and varies by person.

1. Harvard T.H. Chan, The Nutrition Source: Energy Drinks
2. American Heart Association: Added Sugars
3. Liu & Xu, Hereditas 2026: caffeine absorption, half-life, smokers, pills, pregnancy
4. Arnaud, Pharmacokinetics of methylxanthines (stomach vs intestine)
5. Porkka-Heiskanen et al., Science 1997 (adenosine builds while awake)
6. Svatikova et al., JAMA 2015 (Mayo Clinic, blood pressure)
7. Shah et al., J Am Heart Assoc 2019 (32 oz, heart rhythm)
8. Cleveland Clinic: Norepinephrine
9. Mantantzis et al. 2019, Neurosci Biobehav Rev (sugar and mood)
10. NIH Office of Dietary Supplements: Riboflavin
11. Drake et al., J Clin Sleep Med 2013 (AASM, ScienceDaily)
12. FDA: Spilling the Beans
13. SAMHSA: DAWN report on energy drinks
14. Pharmaoffer: bulk caffeine prices (12 months to Aug 2026)
15. Company full-year 2025 results, SEC filing (Feb 2026)

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